

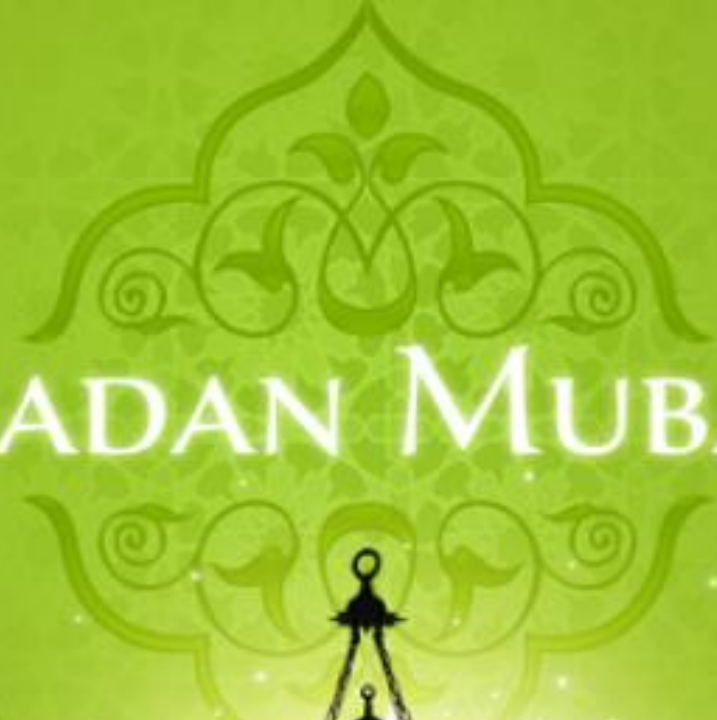


Signals and Systems EE252, Spring 2026

Instructor: Tariq Mumtaz

Semester week # 07





RAMADAN MUBARAK



Islamic Year –1447
19th Feb – 20th March 2026
Spring2026: Wk7 - 10

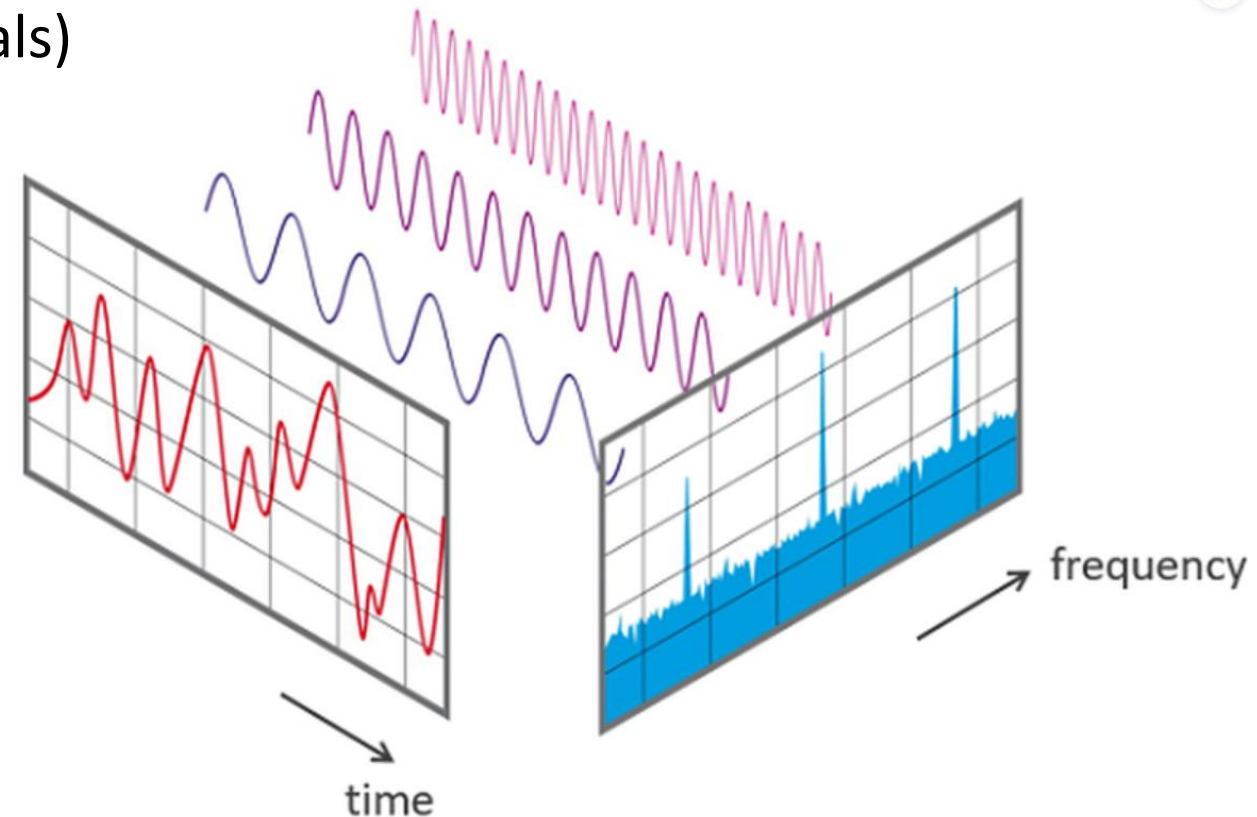


Course Learning Outcome (CLOs):

CLO-1	Apply knowledge of signals and systems to categorize and perform basic operations on signals and systems.	PLO-1 (Engineering Knowledge)	Cognitive Level-3 (Applying)
CLO-2	Calculate the response of DT and CT LTI systems to various inputs and categorize LTI systems.	PLO-1 (Engineering Knowledge)	Cognitive Level-3 (Applying)
CLO-3	Analyze DT signals and systems in the transform domain using the Z transform.	PLO-2 (Problem Analysis)	Cognitive Level-4 (Analyzing)
CLO-4	Analyze CT signals and systems in the transform domain using Laplace, Fourier transforms, and Fourier Series.	PLO-2 (Problem Analysis)	Cognitive Level-4 (Analyzing)
CLO-5	Investigate complex engineering problems in signals and systems and formally present the results.	PLO-4 (Investigation)	Cognitive Level-5 (Evaluating)

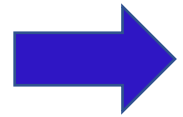
Learning outcome current week sessions:

- Time domain vs. Frequency domain analysis
- Need for Transforms
 - Fourier Transform (CT and DT Signals)
 - Laplace Transform (CT systems)
 - Z Transform (DT Systems)
- The complex exponential signal



Announcements:

HW2 posted 16th Feb – due date 23rd Feb 2026



Quiz 2 date 25th Feb 2026 (Chapter 3, DT Signals & Systems)

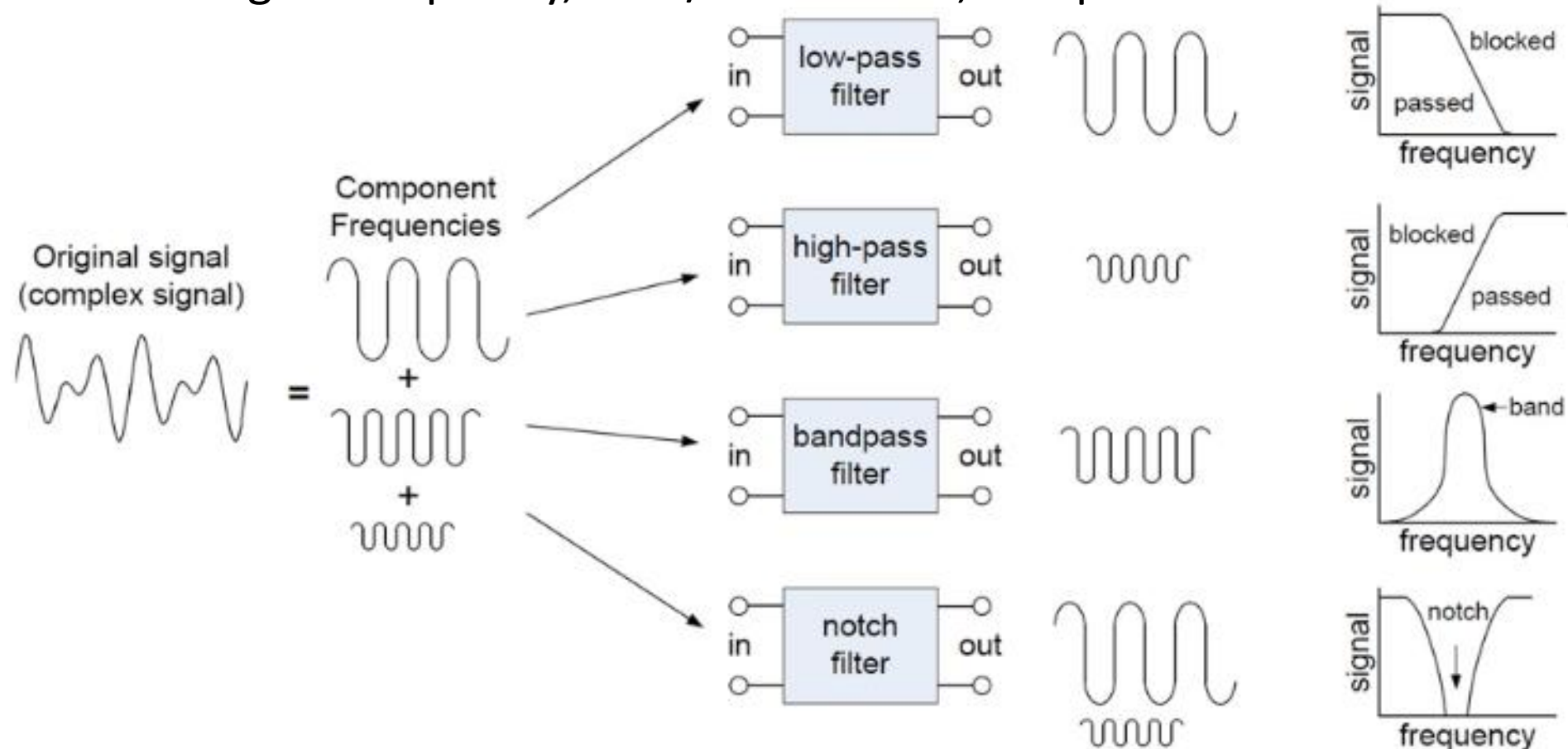
Midterm Exam – 9th Week of Spring 2026 (Chapters 1 & 3)



Time domain Vs. Frequency domain

Time domain Vs. Frequency domain

The frequency domain helps to analyze signals and systems, specifically in filter design – main parameters: signal frequency, Gain/attenuation, and phase



Time domain Vs. Frequency domain

Example: A DT differentiator in the time domain is a high-pass filter in the frequency domain.

$$y[n] = x[n] - x[n - 1]$$

$$h[n] = \delta[n] - \delta[n - 1]$$

For LTI systems, test with:

$$x[n] = e^{j\omega n}$$

Now compute output using convolution:

$$y[n] = x[n] * h[n]$$

Since:

$$h[n] = \delta[n] - \delta[n - 1]$$

Convolution gives:

$$y[n] = x[n] - x[n - 1]$$

Substitute exponential:

$$y[n] = e^{j\omega n} - e^{j\omega(n-1)}$$

Factor:

$$y[n] = e^{j\omega n} (1 - e^{-j\omega})$$

Magnitude of frequency response:

$$|1 - e^{-j\omega}| = 2 \left| \sin \left(\frac{\omega}{2} \right) \right|$$

- At $\omega = 0$:

$$|y[n]| = 0$$

Low frequencies suppressed.

- At $\omega = \pi$:

$$|y[n]| = 2$$

High frequencies amplified.

Time domain Vs. Frequency domain

$$|1 - e^{-j\omega}| = 2 \left| \sin \left(\frac{\omega}{2} \right) \right|$$

$$e^{-j\omega} = \cos \omega - j \sin \omega$$

$$1 - e^{-j\omega} = 1 - \cos \omega + j \sin \omega$$

$$|1 - e^{-j\omega}| = \sqrt{(1 - \cos \omega)^2 + (\sin \omega)^2}$$

$$= \sqrt{1 - 2 \cos \omega + \cos^2 \omega + \sin^2 \omega}$$

$$= \sqrt{2(1 - \cos \omega)}$$

$$1 - \cos \omega = 2 \sin^2(\omega/2)$$

$$= \sqrt{4 \sin^2(\omega/2)}$$

$$\boxed{|1 - e^{-j\omega}| = 2 |\sin(\omega/2)|}$$



Time domain Vs. Frequency domain

Example: A DT integrator (or averaging filter) in the time domain is a low-pass filter in the frequency domain.

$$y[n] = \frac{1}{2}(x[n] + x[n-1])$$

Now apply:

$$h[n] = \frac{1}{2}(\delta[n] + \delta[n-1])$$

Output:

$$x[n] = e^{j\omega n}$$

$$y[n] = \frac{1}{2}(e^{j\omega n} + e^{j\omega(n-1)})$$

Factor:

$$y[n] = e^{j\omega n} \frac{1}{2}(1 + e^{-j\omega})$$

Magnitude:

$$\left| \frac{1}{2}(1 + e^{-j\omega}) \right| = \left| \cos\left(\frac{\omega}{2}\right) \right|$$

- At $\omega = 0$:

Maximum value

$$|y[n]| = 1$$

- At $\omega = \pi$:

$$|y[n]| = 0$$

NOTE: Testing signal for the frequency response $x[n]$ is complex Sinusoid ----- why????



Need for Transforms in Signals & Systems Analysis

Need for Transforms in Signal & Systems Analysis:

Importance of the Transforms – 01

Transforms provide frequency-domain insight

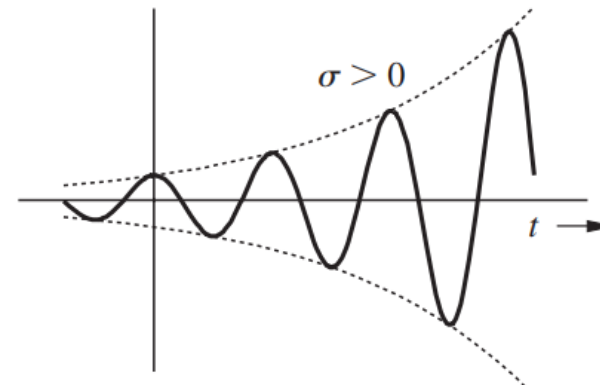
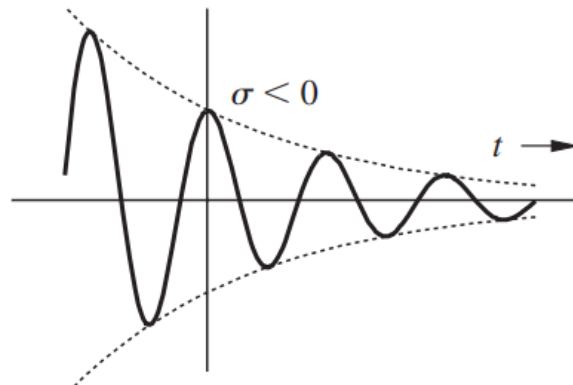
$$\frac{dy(t)}{dt} + ay(t) = x(t)$$

t: time domain

Laplace Transform

$$sY(s) + aY(s) = X(s)$$

S : s ($\sigma + j\omega$) domain





Need for Transforms in Signal & Systems Analysis:

Importance of the Transforms - 02

Convert convolution into multiplication

$$y(t) = x(t) * h(t)$$



$$Y(s) = X(s)H(s)$$



Need for Transforms in Signal & Systems Analysis:

Importance of the Transforms - 03

Transform converts differential (continuous time)/ difference equations (discrete time) into algebraic equations. For example

$$\frac{dy(t)}{dt} + ay(t) = x(t)$$

Laplace Transform

$$sY(s) + aY(s) = X(s)$$



Need for Transforms in Signal & Systems Analysis:

Importance of the Transforms - 03

Without transforms, to analyze any systems represented as differential or difference equations, we have to

- Solve homogeneous equation
- Apply initial conditions manually
- Combine natural (Initial condition, zero input response) and forced response (zero state response)

But with transforms:

- Everything becomes algebraic
- Initial conditions automatically appear during transformation
- Solution becomes a straightforward substitution



Important Transform Used in Signals & Systems Analysis



Need for Transforms in Signal & Systems Analysis:

- Transforms help to attain a compact mathematical description of Signals & Systems using complex exponentials. Any signal/system can be expressed as a combination of complex sinusoids/ complex exponentials.
- Transforms allow decomposition of the signal into complex sinusoids ($e^{j\omega}$)
- Transforms also allow system representation in exponential form ($e^{\sigma+j\omega}$)



Need for Transforms in Signal & Systems Analysis:

Transforms # 1

Continuous-Time Fourier Transform (CTFT):

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$$

$$e^{j\omega t} = \cos(\omega t) + j \sin(\omega t)$$

The signal is decomposed into infinitely many complex sinusoids. So the Fourier Transform tells us:

- I. What frequencies are present
- II. How strong they are

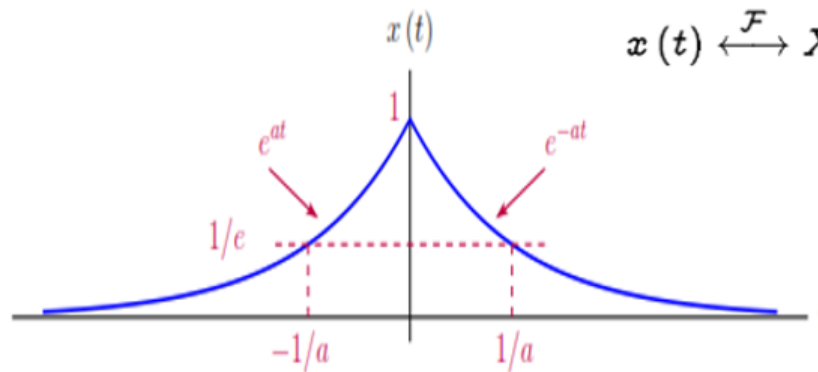
Need for Transforms in Signal & Systems Analysis:

Transforms # 1

Continuous-Time Fourier Transform (CTFT):

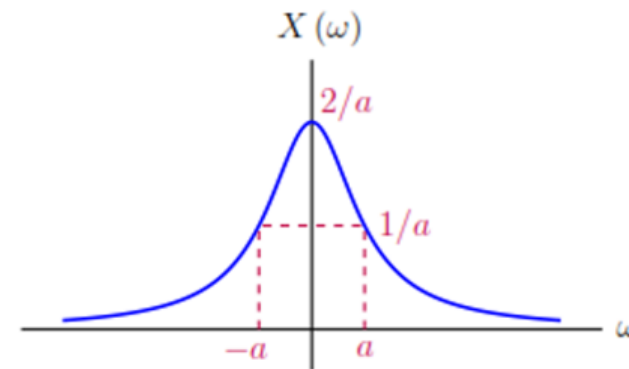
$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$$

Exponential Decaying signal



Time Domain

$$x(t) \xleftrightarrow{\mathcal{F}} X(\omega)$$



Frequency Domain
(Spectrum)

Need for Transforms in Signal & Systems Analysis:

Transforms # 2

Discrete-Time Fourier Transform (DTFT):

$$X(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega n}$$

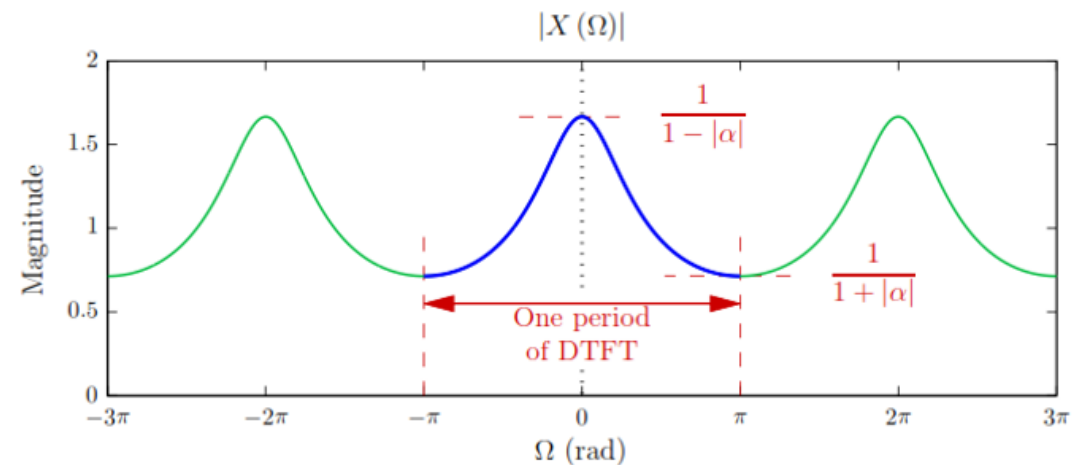
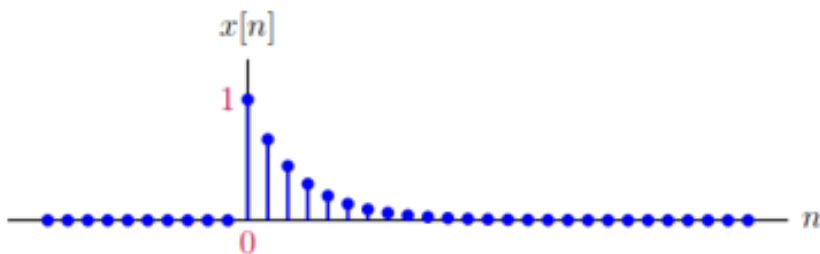
$$\Omega = \omega T_s$$

ω : radians/sec

Ω : radians/sample

T_s : Sampling period

Again, the signal is represented as a sum of complex sinusoids.





Need for Transforms in Signal & Systems Analysis:

Advantages of Exponential Representation

$$\frac{d}{dt}e^{j\omega t} = j\omega e^{j\omega t}$$

Differentiation becomes multiplication by $j\omega$, a major simplification that converts differential equations into algebraic equations



Need for Transforms in Signal & Systems Analysis:

Transforms # 3

Laplace Transform (Continuous-Time Systems)

$$X(s) = \mathcal{L}\{x(t)\} = \int_{-\infty}^{\infty} x(t) e^{-st} dt$$

The signal is expressed in terms of complex exponentials e^{st}

Here, $S = \sigma + j\omega$ (Complex exponential) to represent exponentially growing/decaying complex sinusoids.

Unlike CTFT (which uses $e^{j\omega t}$), Laplace allows growth/decay ($e^{\sigma t}$) and sinusoidal oscillation ($e^{j\omega t}$) simultaneously.



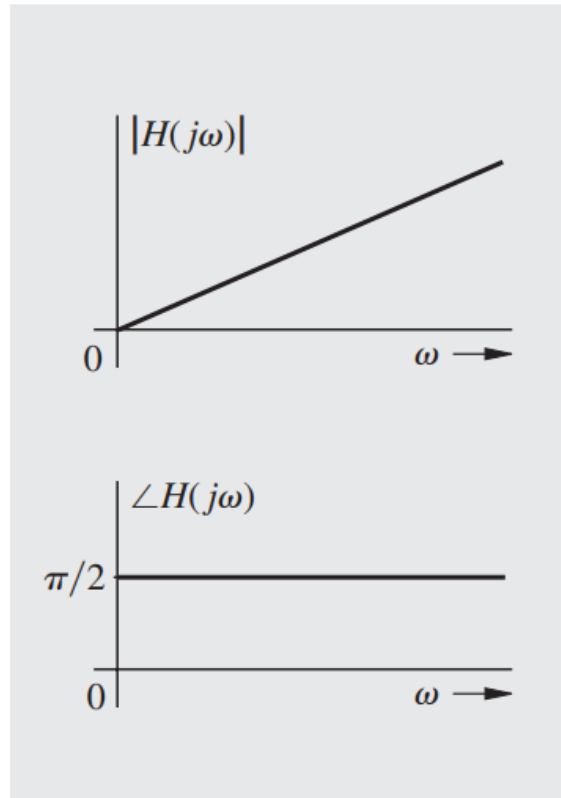
Need for Transforms in Signal & Systems Analysis:

Transforms # 3

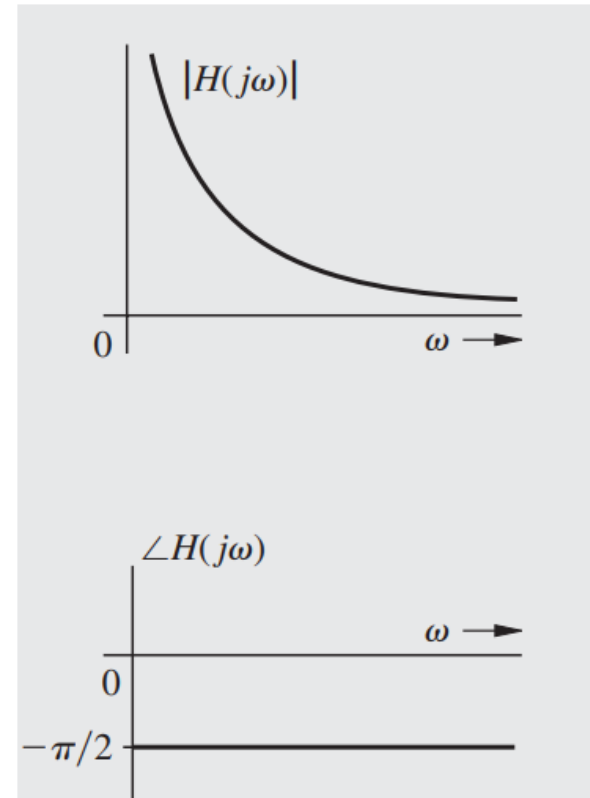
Laplace Transform (Continuous-Time Systems)

$$X(s) = \mathcal{L}\{x(t)\} = \int_{-\infty}^{\infty} x(t) e^{-st} dt$$

Ideal Differentiator



Ideal Integrator





Need for Transforms in Signal & Systems Analysis:

Transforms # 4

Z Transform (Discrete-Time Systems)

$$X(z) = \mathcal{Z}\{x[n]\} = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

Here $z=re^{j\omega}$ is a complex exponential. Just like DTFT represents signals as a sum of complex exponentials $e^{j\omega n}$, the Z-transform generalizes this with z^{-n} , allowing both oscillation and growth/decay.

For $|z|=1$, Z-transform reduces to DTFT:

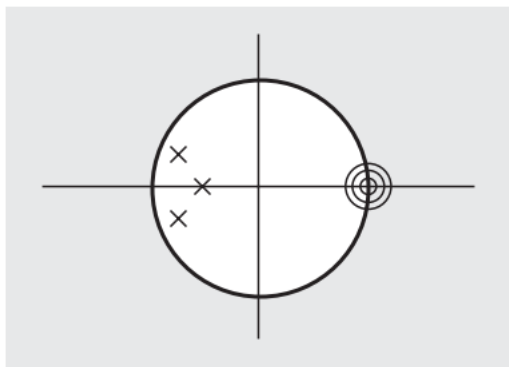
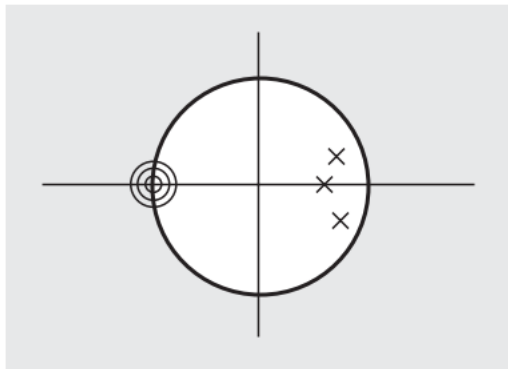
$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

Need for Transforms in Signal & Systems Analysis:

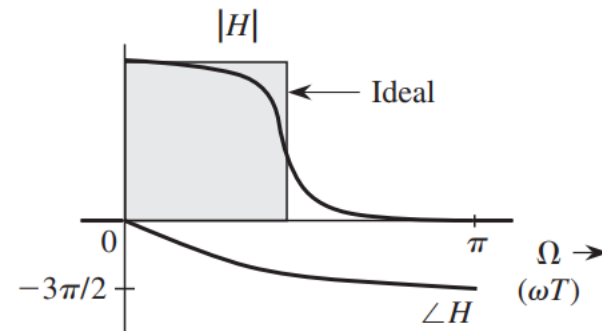
Transforms # 4

Z Transform (Discrete-Time Systems)

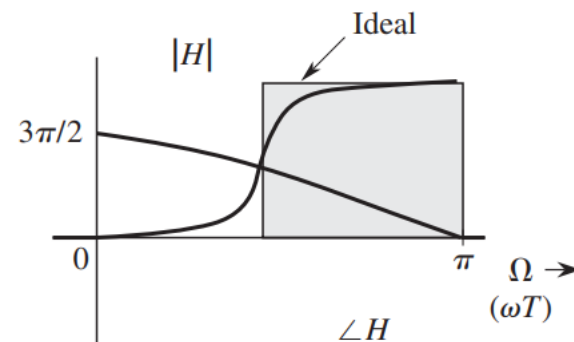
$$X(z) = \mathcal{Z}\{x[n]\} = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$



(d)



(e)





Need for Transforms in Signal & Systems Analysis:

- CTFT/DTFT: decomposition into pure complex exponentials $e^{j\omega t}$ or $e^{j\omega n}$.
- Laplace/Z-Transform: decomposition into general complex exponentials e^{st} or z^n
- CTFT and DTFT are primarily used for spectral (frequency) analysis, while Laplace and Z-transforms are more powerful for system analysis because they include stability/causality and transient behavior.

$$\text{CTFT} = X(s) \Big|_{s=j\omega}$$

$$x(t) \xleftrightarrow{\text{CTFT}} X(\omega)$$

$$H(\omega) = \text{CTFT of } h(t)$$

$$x[n] \xleftrightarrow{\text{DTFT}} X(e^{j\Omega})$$

$$H(e^{j\Omega}) = \text{DTFT of } h[n]$$

$$\text{DTFT} = X(z) \Big|_{z=e^{j\Omega}}$$



Complex Exponential Signal Continuous Time (CT)

Useful Signals Models:



COMPLEX SINUSOID

$$e^{j\omega t}$$



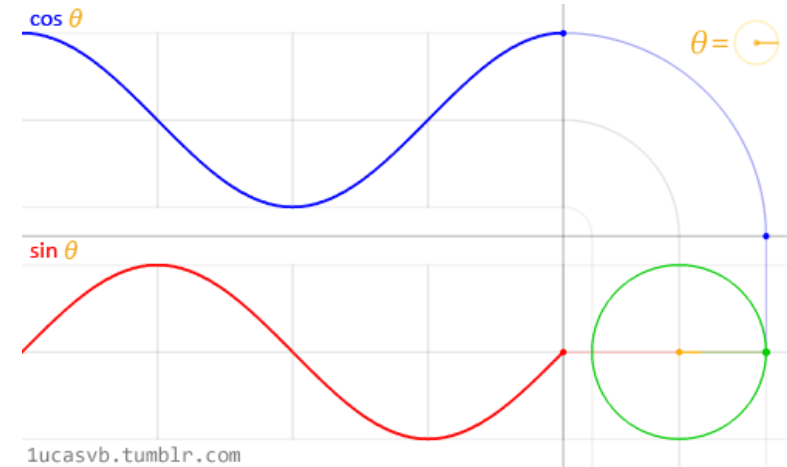
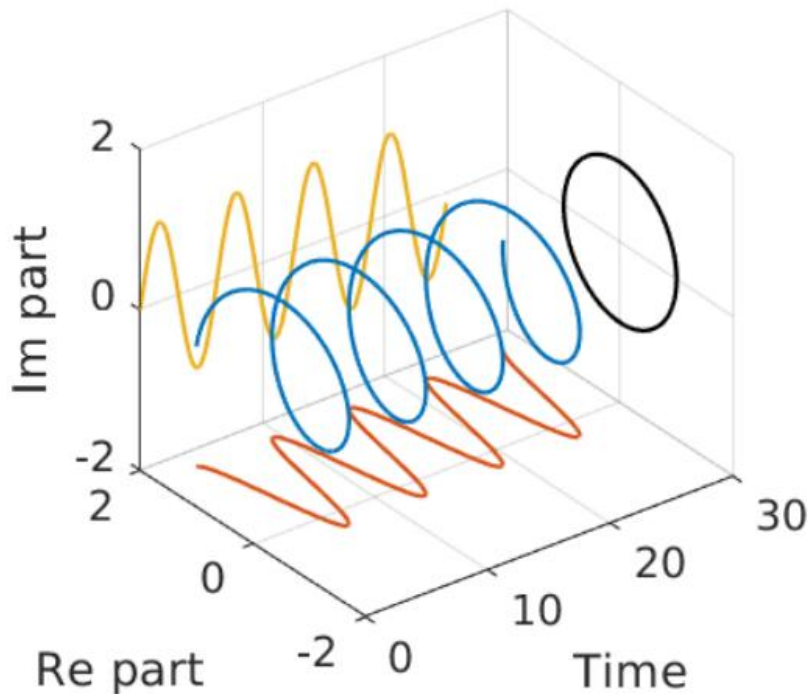
COMPLEX EXPONENTIAL

$$e^{st} = e^{(\sigma + j\omega)t}$$

Useful Signals Models:

COMPLEX SINUSOID

$$e^{j\omega t}$$



- The complex sinusoid is a spiral, whose real and imaginary parts trace cosine and sine.
- The complex sinusoid traces a unit circle in the complex plane as its modulus is always 1.
- What if we allow the modulus to also grow or decrease with time? – need **Complex Exponential!**

Useful Signals Models:

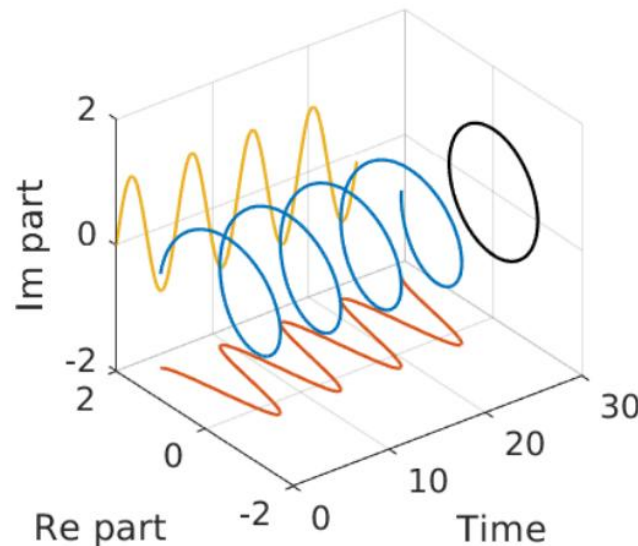
COMPLEX EXPONENTIAL

$$e^{st} = e^{(\sigma + j\omega)t}$$

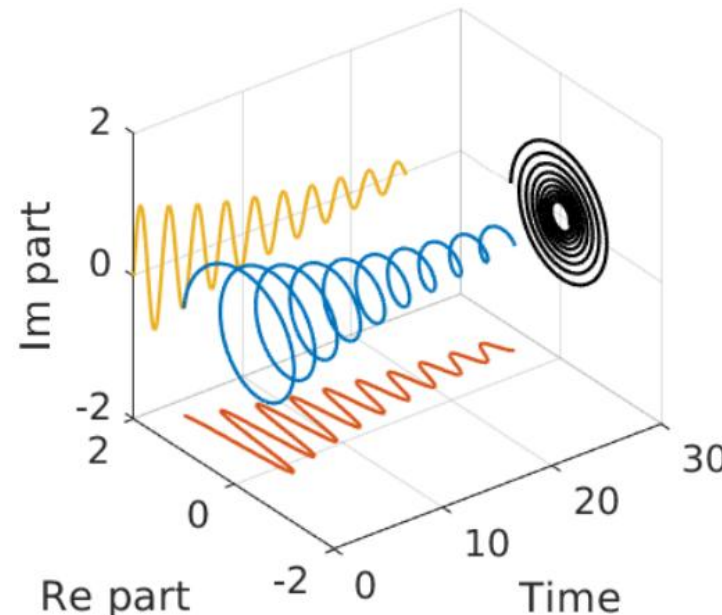
$$e^{(\sigma + j\omega)t} = e^{\sigma t} e^{j\omega t}$$

Complex Exponential is basically a complex Sinusoid with possibly decaying or growing amplitude.

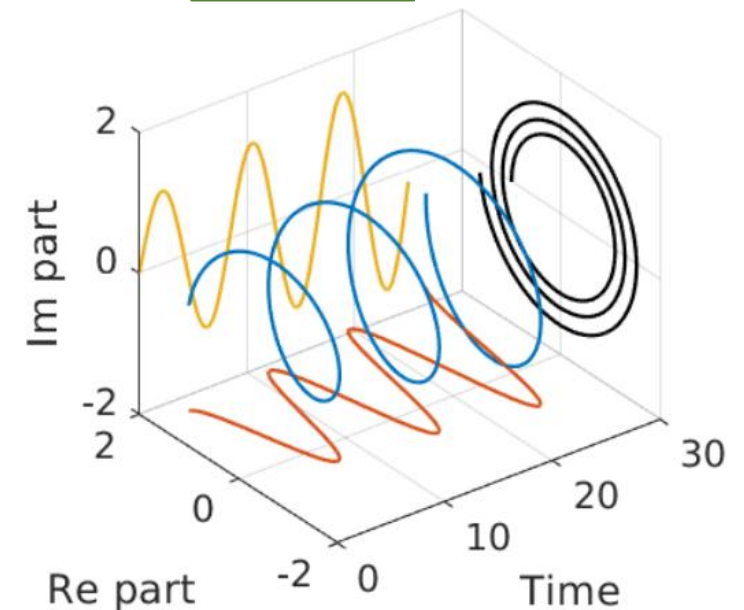
$$\sigma = 0$$



$$\sigma < 0$$



$$\sigma > 0$$





Useful Signals Models:

COMPLEX SINUSOID

$$e^{j\omega t}$$



Periodic with period $T_0 = \frac{2\pi}{\omega}$

COMPLEX EXPONENTIAL

$$e^{st} = e^{(\sigma + j\omega)t}$$



Not periodic, unless $\sigma = 0$

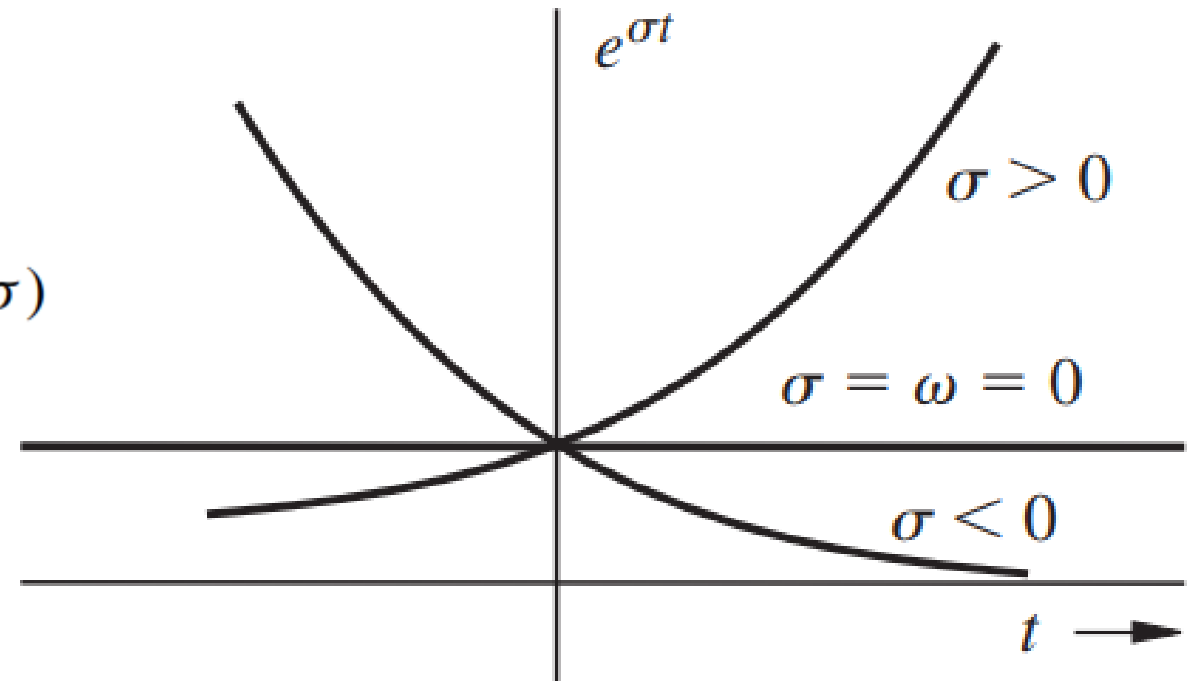
Useful Signals Models:

The exponential function e^{st}

The function e^{st} encompasses a large class of functions, for example

$$s = \sigma + j\omega$$

1. A constant $k = ke^{0t}$ ($s = 0$)
2. A monotonic exponential $e^{\sigma t}$ ($\omega = 0, s = \sigma$)

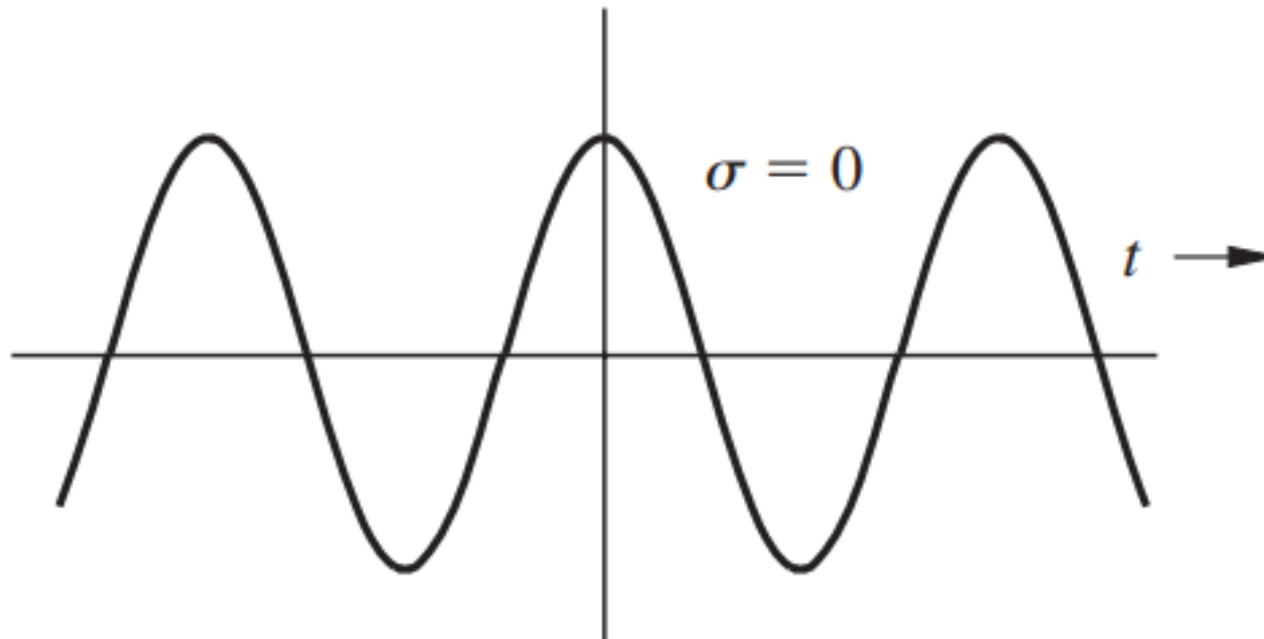


Useful Signals Models:

The exponential function e^{st}

$$s = \sigma + j\omega$$

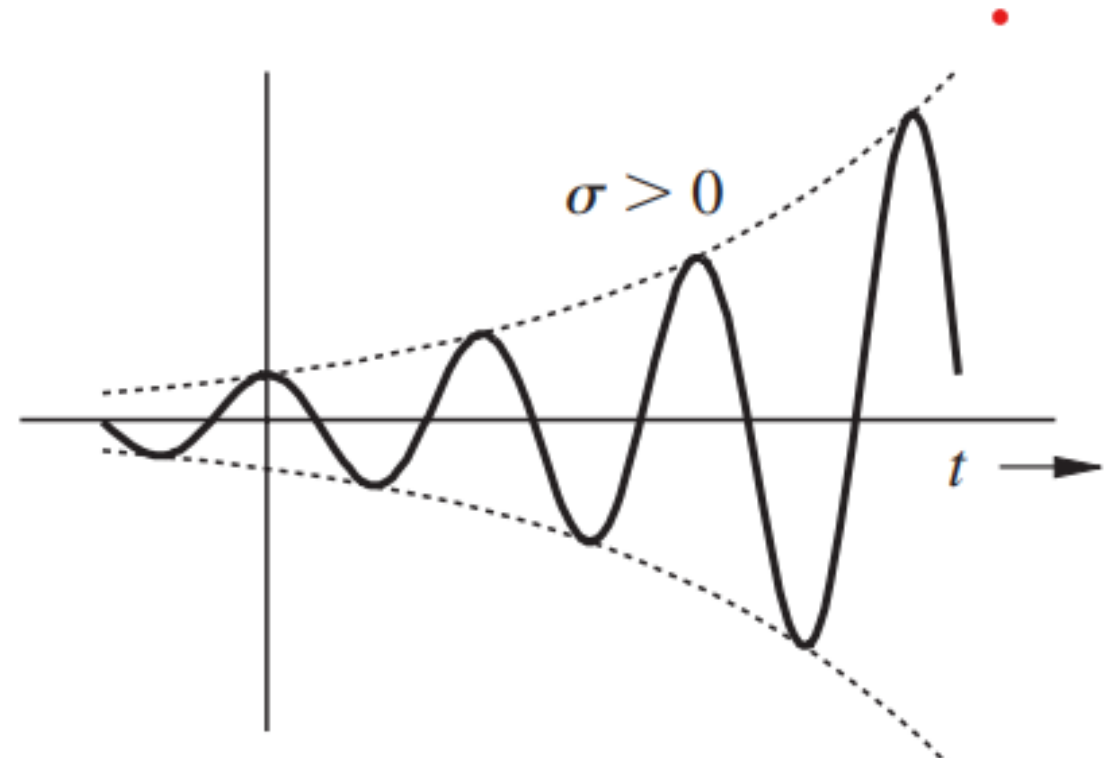
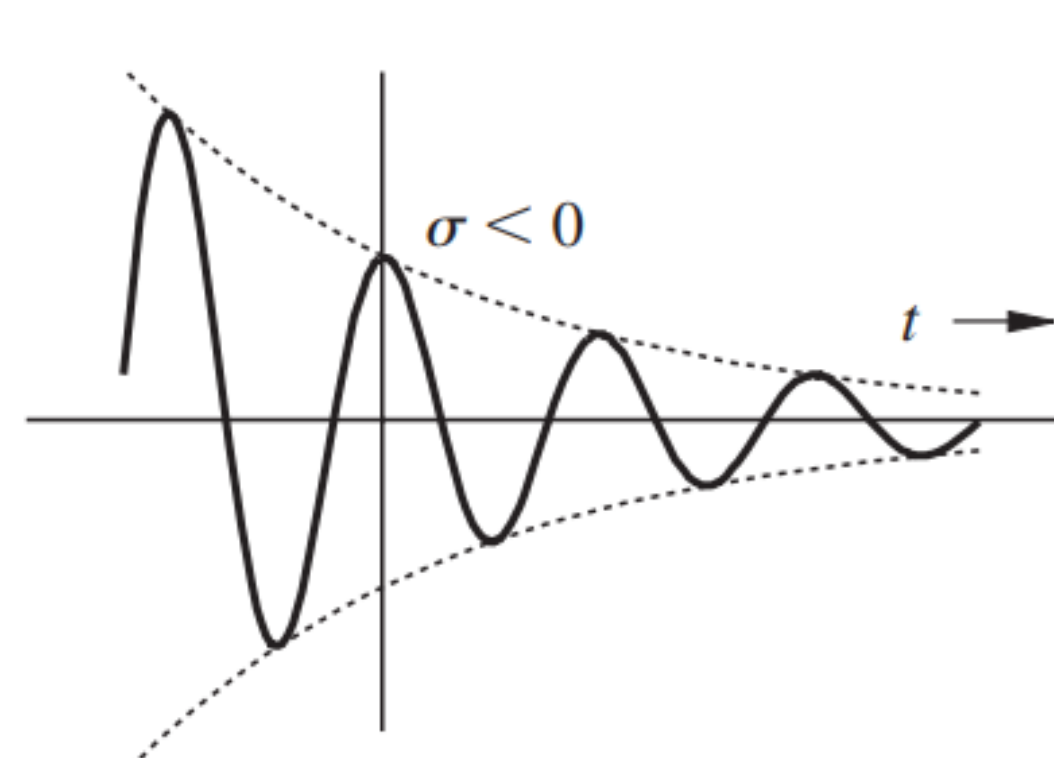
3. A sinusoid $\cos \omega t$ ($\sigma = 0, s = \pm j\omega$)



Useful Signals Models:

The exponential function e^{st}

4. An exponentially varying sinusoid $e^{\sigma t} \cos \omega t$ ($s = \sigma \pm j\omega$)





Useful Signals Models:

The exponential function e^{st}

The complex frequency “ s ” is represented on a complex frequency plane (s plane). The s plane is separated into two parts:

- The left half-plane (LHP) corresponds to exponentially decaying signals
- The right half-plane (RHP) corresponds to exponentially growing signals.

The imaginary axis separates the two regions and corresponds to signals of constant amplitude.

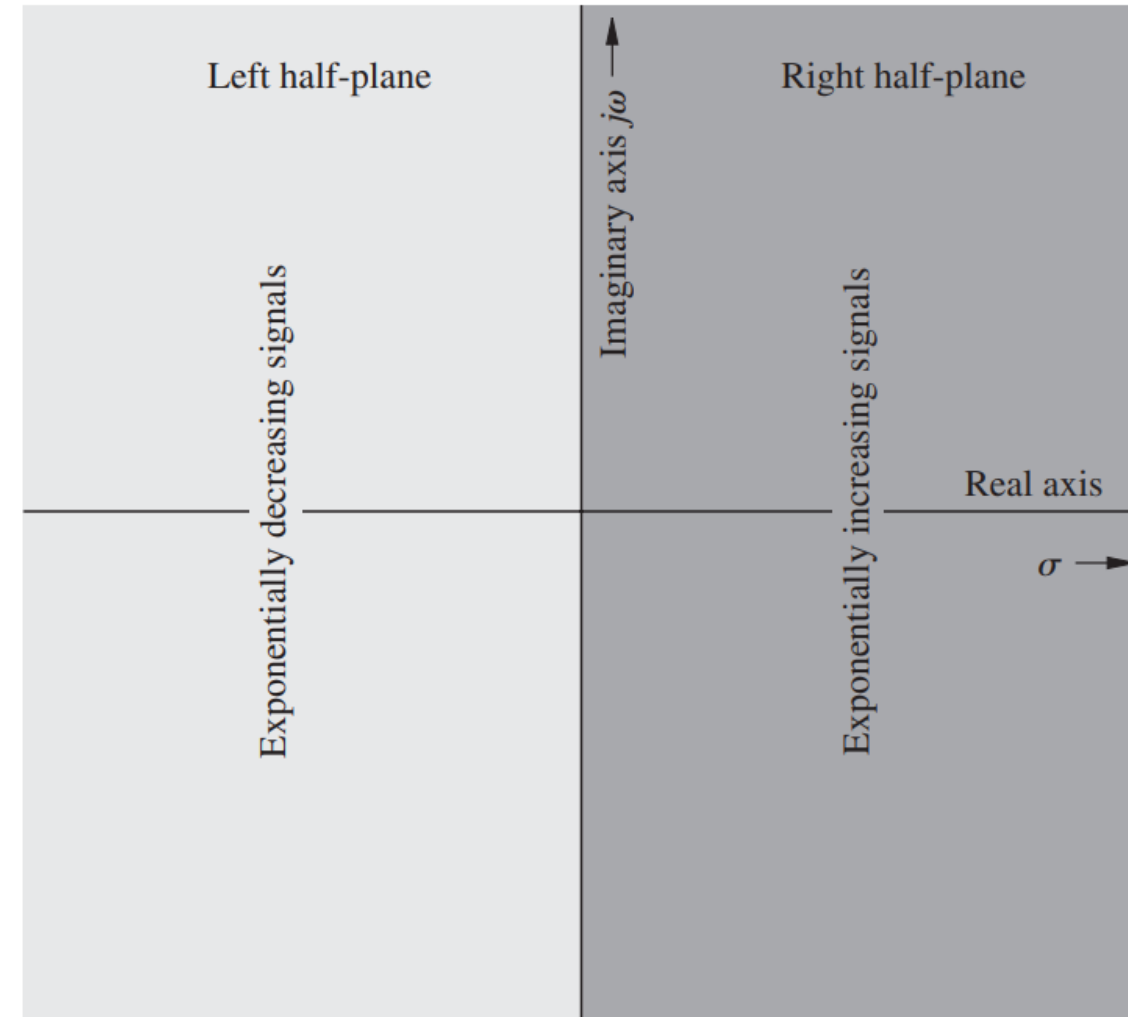


Figure 1.22 Complex frequency plane.



Useful Signals Models:

The exponential function e^{st}

An exponentially growing sinusoid $e^{2t} \cos 5t$, for example, can be expressed as a linear combination of exponentials $e^{(2+j5)t}$ and $e^{(2-j5)t}$ with complex frequencies $2 + j5$ and $2 - j5$, respectively, which lie in the RHP.

Working:
$$e^{\sigma t} \cos \omega t = \frac{1}{2}(e^{st} + e^{s^*t})$$

$$s = \sigma + j\omega$$

For given, $e^{2t} \cos 5t \Rightarrow$ here $\sigma = 2$ and $\omega = 5$
therefore, $s = 2 + j5$ and $s^* = 2 - j5$

Cos (ωt) – a real signal (not imaginary) must have a conjugate complex frequency pair

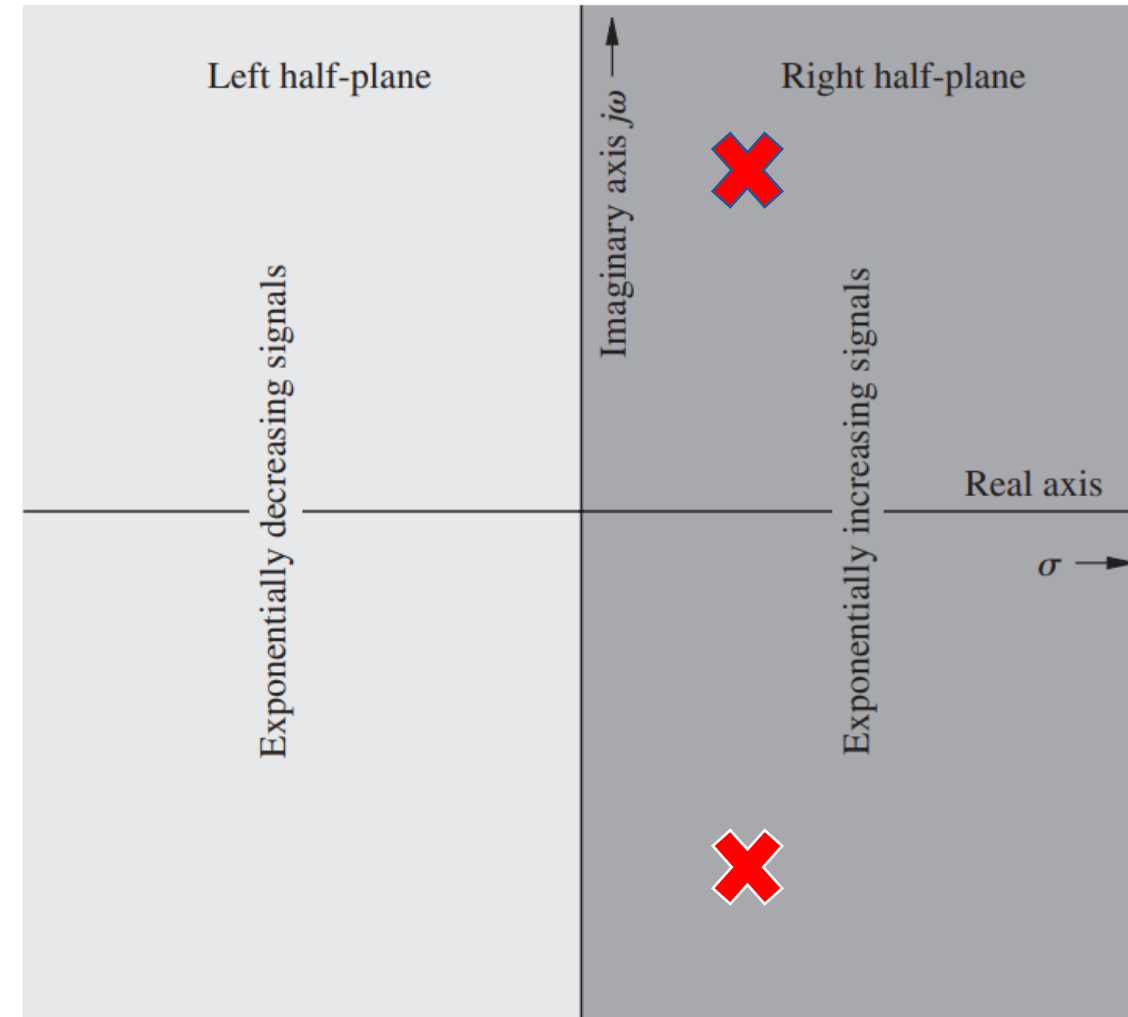


Figure 1.22 Complex frequency plane.

Useful Signals Models:

The exponential function e^{st}

An exponentially growing sinusoid $e^{2t} \cos 5t$, for example, can be expressed as a linear combination of exponentials $e^{(2+j5)t}$ and $e^{(2-j5)t}$ with complex frequencies $2 + j5$ and $2 - j5$, respectively, which lie in the RHP.

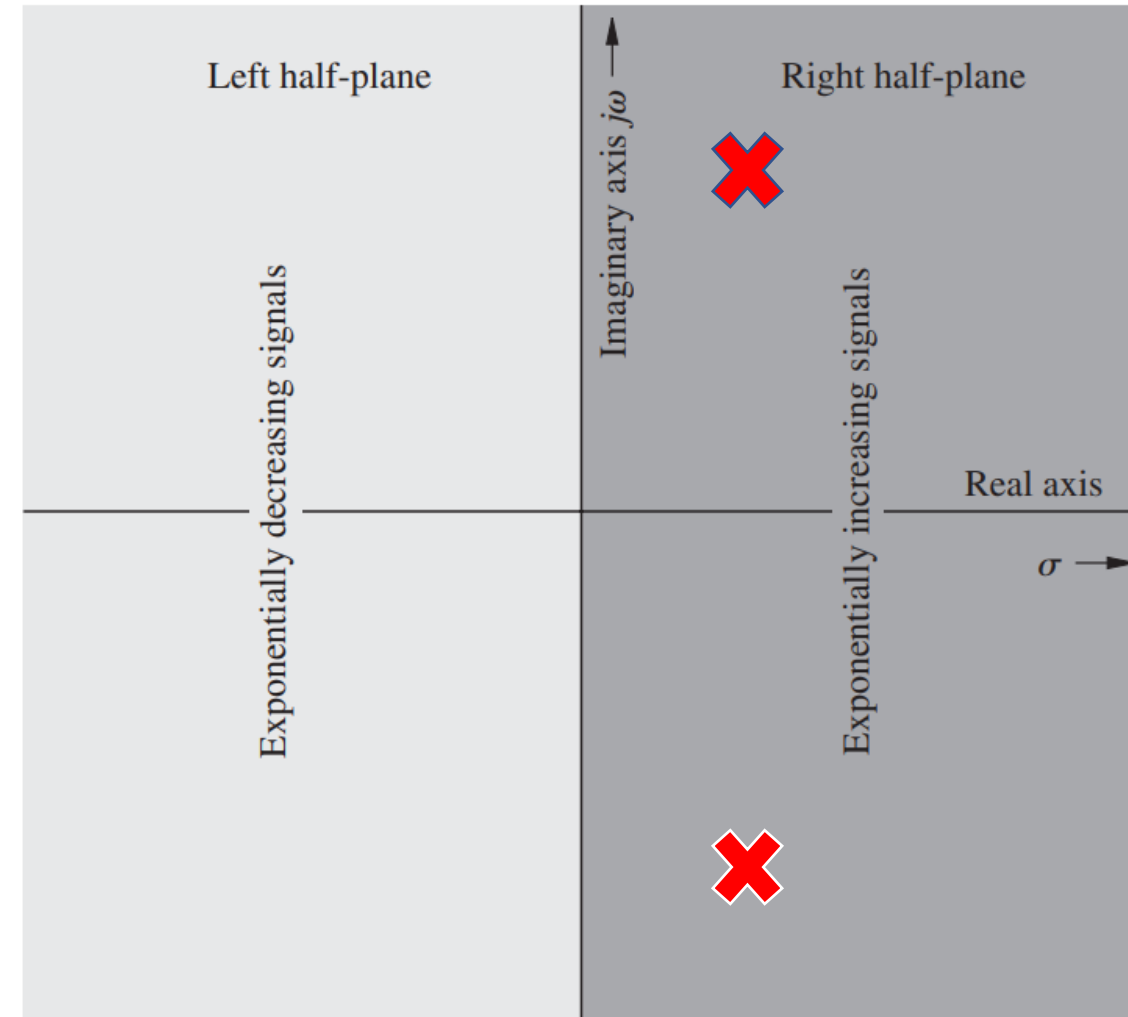
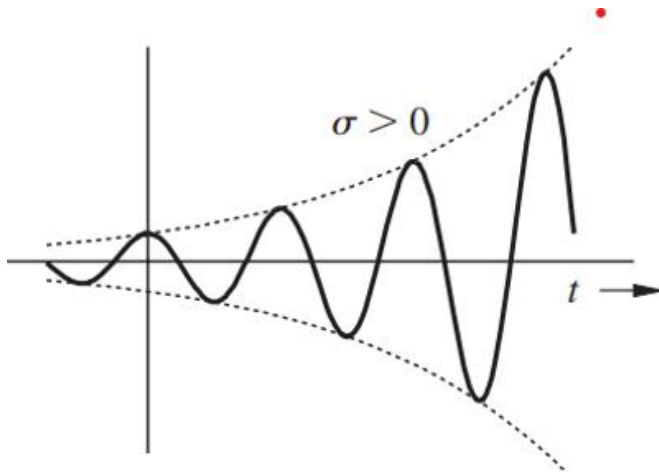


Figure 1.22 Complex frequency plane.

Useful Signals Models:

The exponential function e^{st}

An exponentially decaying sinusoid $e^{-2t} \cos 5t$, for example, can be expressed as a linear combination of exponentials $e^{(-2+j5)t}$ and $e^{(-2-j5)t}$ with complex frequencies $2 + j5$ and $2 - j5$, respectively, which lie in the LHP.

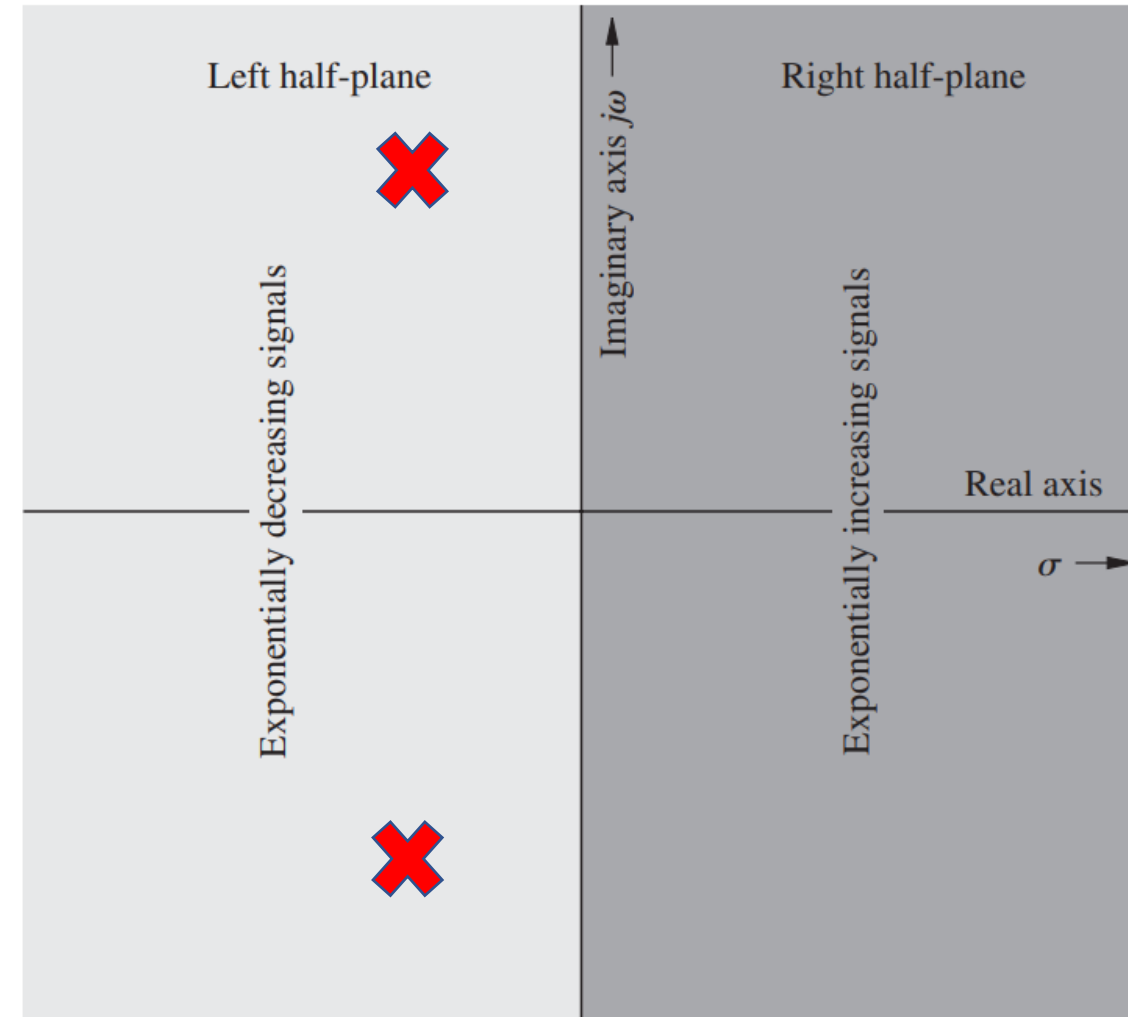
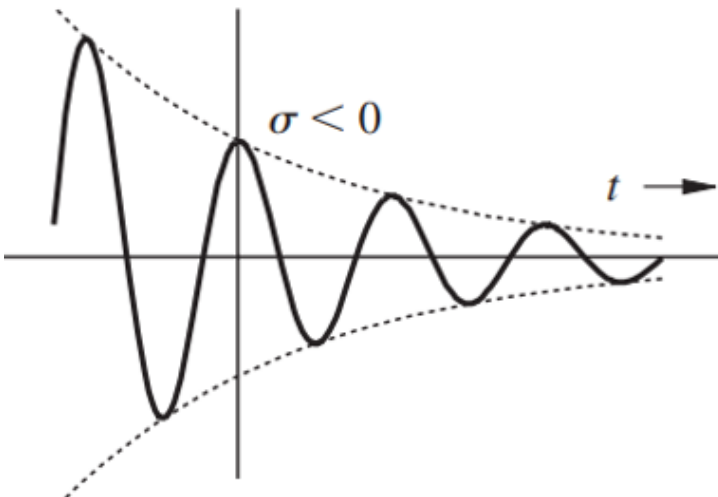


Figure 1.22 Complex frequency plane.

Useful Signals Models:

The exponential function e^{st}

A constant-amplitude sinusoid $\cos 5t$ can be expressed as a linear combination of exponentials e^{j5t} and e^{-j5t} with **complex frequencies $\pm j5$** , which lie on the imaginary axis.

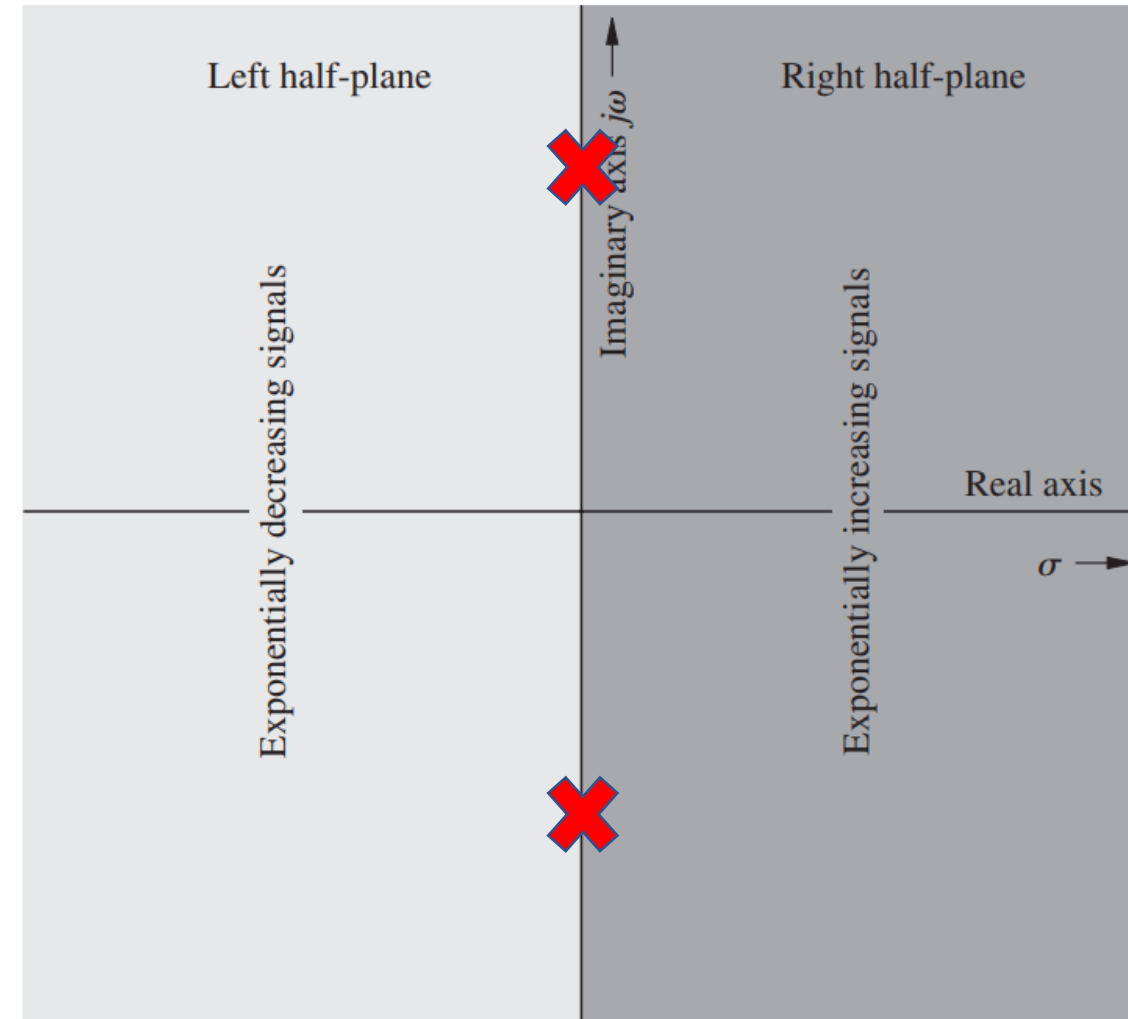
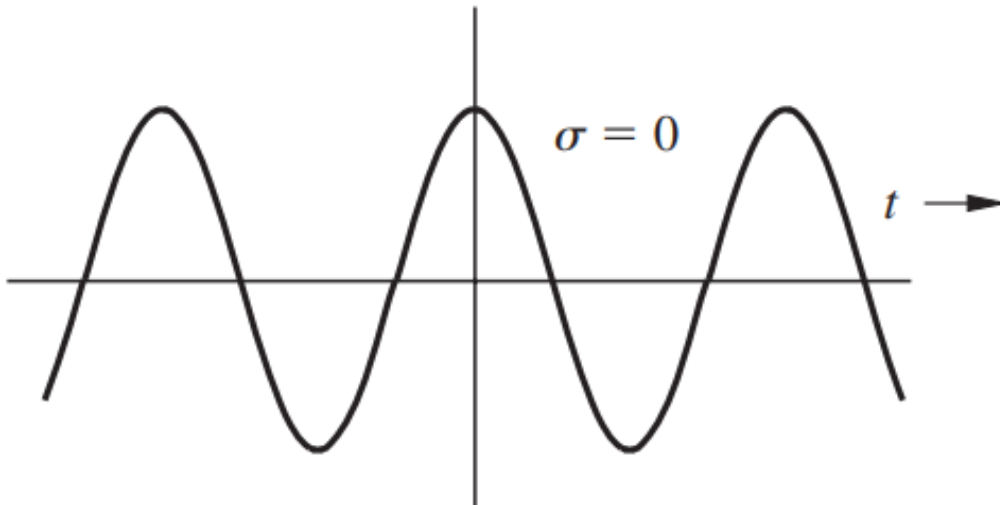


Figure 1.22 Complex frequency plane.



Complex Exponential Signal Discrete Time (DT)



Discrete time (DT) Signals & Systems:

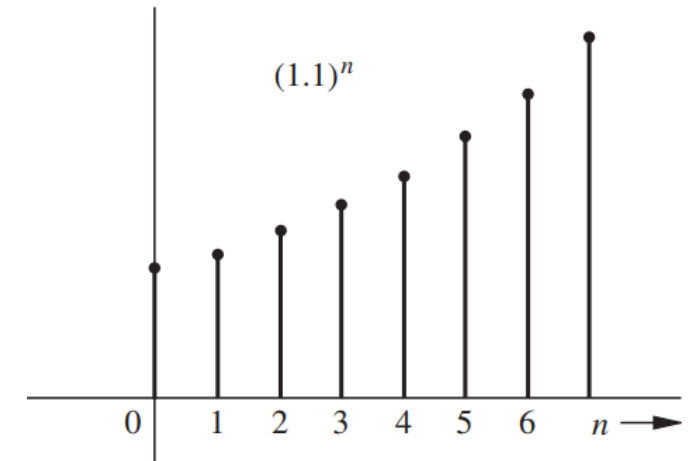
DT Exponential signal - γ^n

A continuous-time exponential signal is of the form:

$$x[n] = e^{an}$$

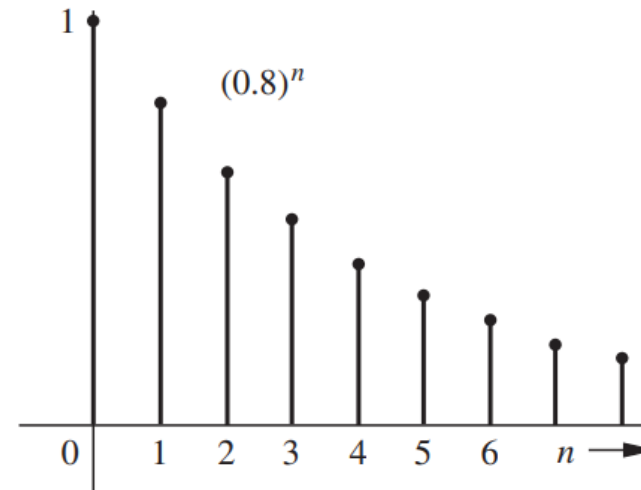
$$\text{let, } \gamma = e^a$$

$$x[n] = \gamma^n$$



Assuming γ is real number

- For $\gamma > 1$, $x[n]$ is growing exponential
- For $0 < \gamma < 1$, $x[n]$ is decaying exponential
- For $\gamma = 1$, $x[n]$ is a constant signal





Discrete time (DT) Signals & Systems:

DT Complex Exponential signal - γ^n

If γ is a complex number, then

$$x[n] = \gamma^n$$

$$\text{let, } \gamma = e^{a+jb}$$

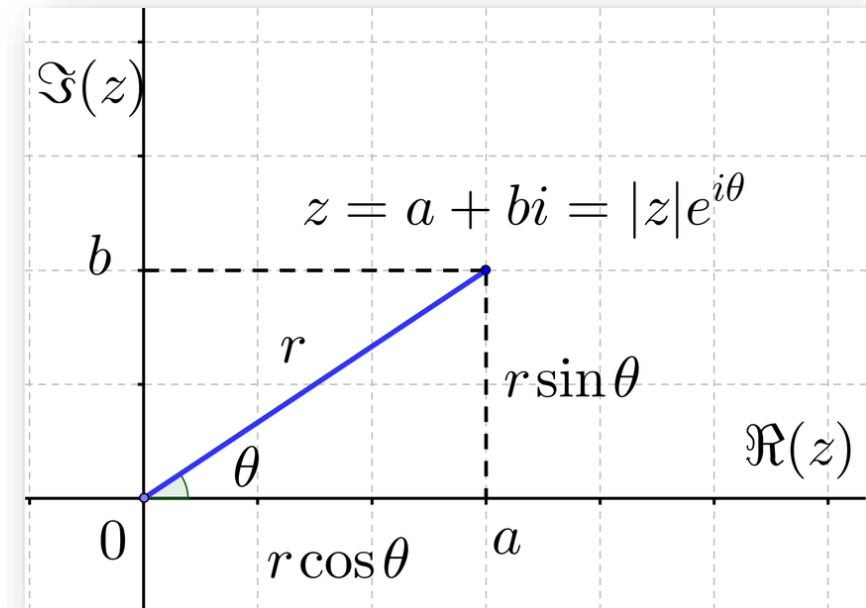
$$x[n] = e^{(a+bj)n}$$

$$\gamma = |\gamma| e^{jb}$$

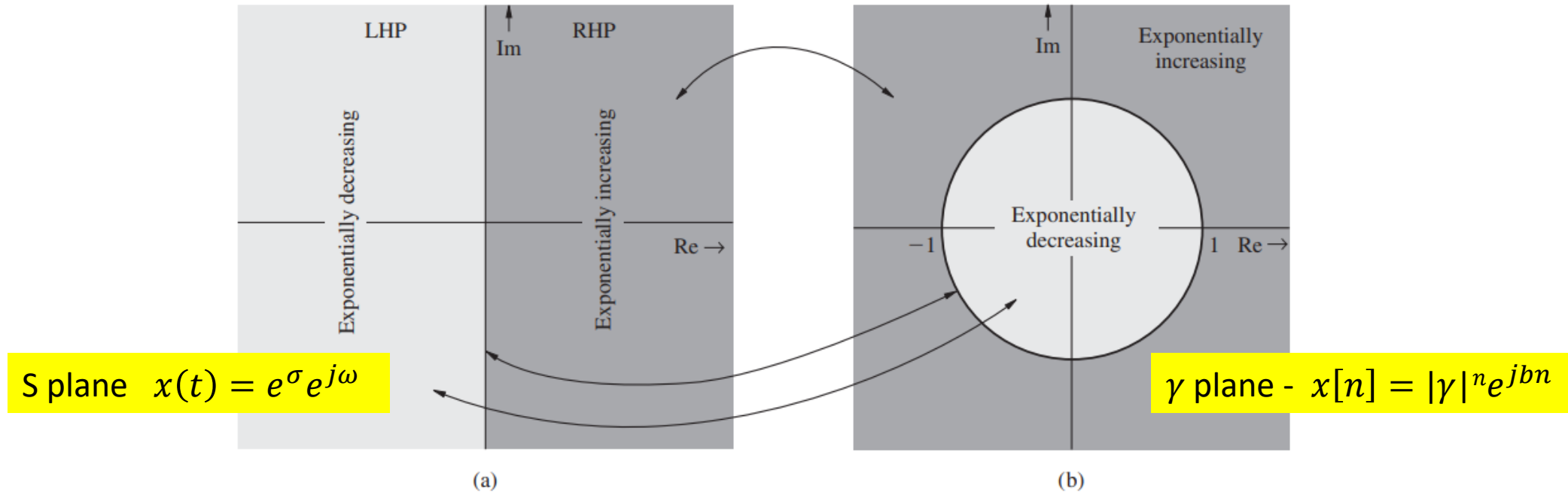
$$\gamma^n = |\gamma|^n e^{jbn}$$

Since the magnitude of e^{jbn} is always unity regardless of the value of n , then the magnitude of γ^n is $|\gamma|^n$

- For $|\gamma| > 1$, $\gamma^n \rightarrow \infty$ as $n \rightarrow \infty$
- For $|\gamma| < 1$, $\gamma^n \rightarrow 0$ as $n \rightarrow \infty$
- For $|\gamma| = 1$, $\gamma^n = 1$ for all n



Discrete time (DT) Signals & Systems:



For $\sigma < 0$: exponentially decreasing

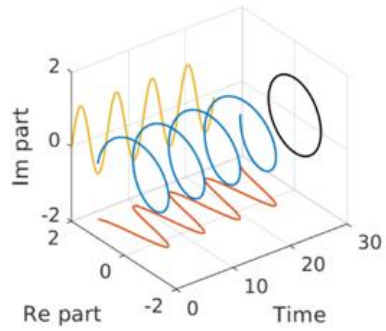
$\sigma > 0$: exponentially increasing

$\sigma = 0$: constant magnitude

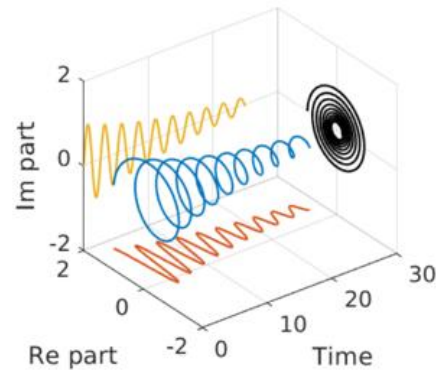
- For $|\gamma| > 1$, $\gamma^n \rightarrow \infty$ as $n \rightarrow \infty$
- For $|\gamma| < 1$, $\gamma^n \rightarrow 0$ as $n \rightarrow \infty$
- For $|\gamma| = 1$, $\gamma^n = 1$ for all n

Discrete time (DT) Signals & Systems:

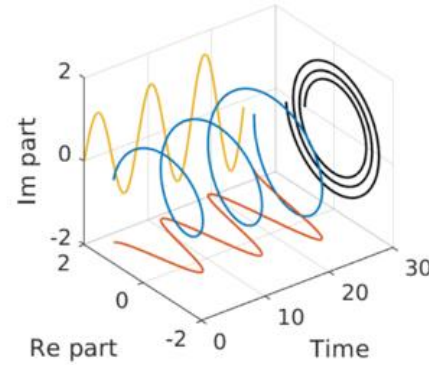
$$\sigma = 0$$



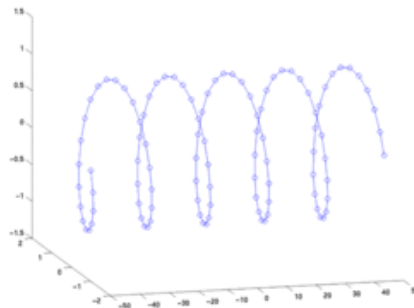
$$\sigma < 0$$



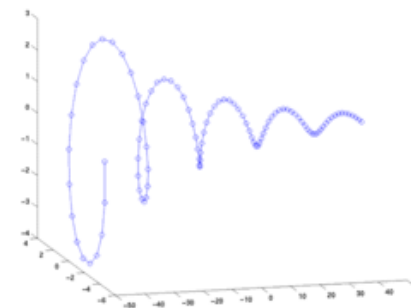
$$\sigma > 0$$



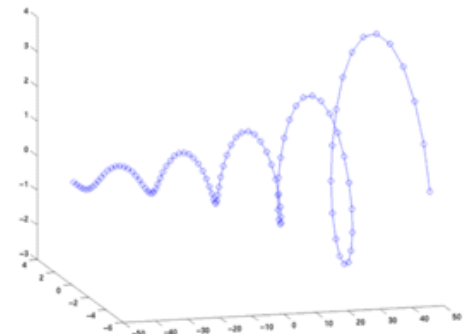
$$|\gamma| = 1$$



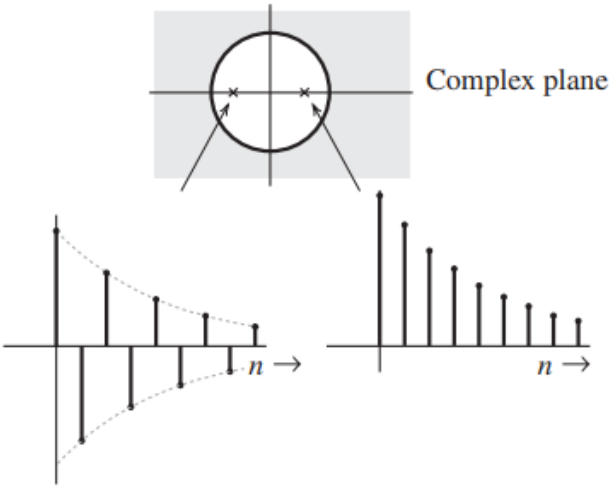
$$|\gamma| < 1$$



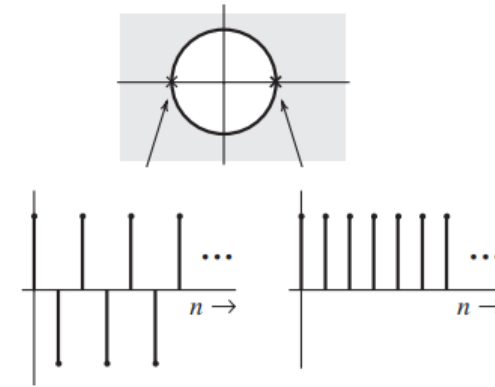
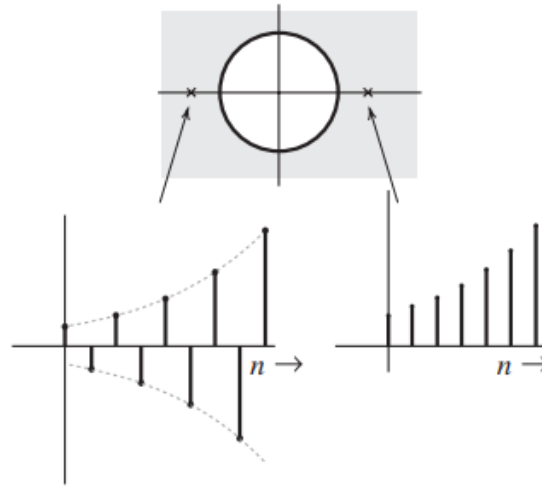
$$|\gamma| > 1$$



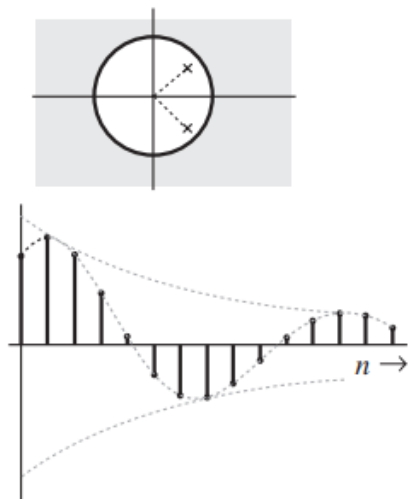
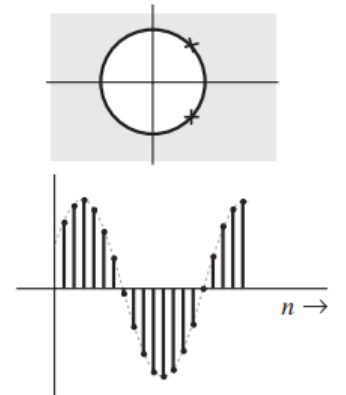
Discrete time (DT) Signals & Systems:



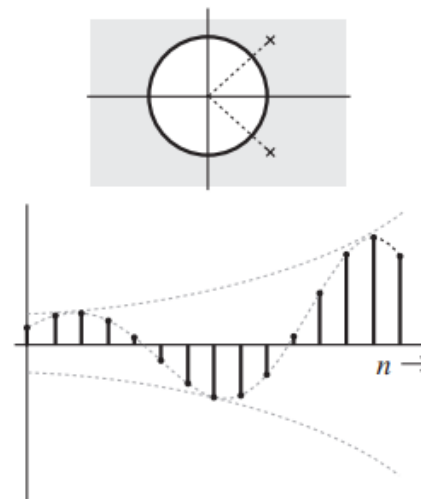
(a)



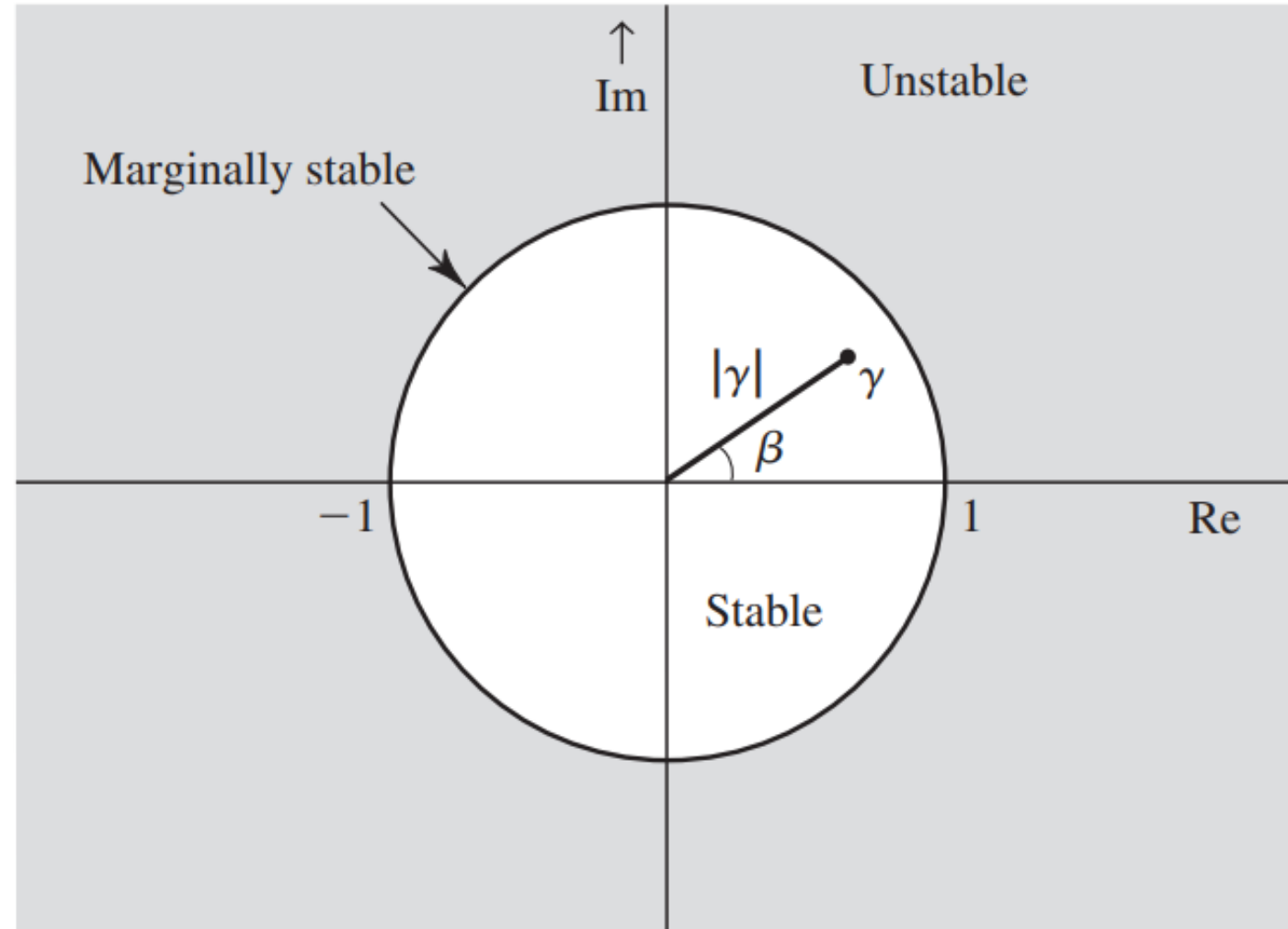
(c)



(b)



Discrete time (DT) Signals & Systems:



Roadmap till Semester end:



Z- Transform (Wk 8 – 10)
Laplace Transform (Wk 11 – 13)
Fourier Transform (Wk 14 – 15)
Project presentations (Wk16)



Session 02- Quiz-II
Thanks!